SinoGNSS®

ComNav Technology

User Guide

M100X GNSS Receiver



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Technical Assistance

If you have any question and can't find the answer in this manual, please contact your local dealer from which you purchased the M100X receiver. Alternatively, request technical support from

ComNav Technology Website: *https://www.comnavtech.com*

or technical support email: *support@comnavtech.com*

Your feedback about this Guide will help us to improve it with future revisions.

Safety Information

Before using the receiver, please make sure that you have read and understood this User Guide, as well as the safety requirements.

- Connect your devices strictly based on this User Guide
- Install the GNSS receiver in a location that minimizes vibration and moisture
- Avoid falling to ground, or colliding with other items
- Do not rotate 26-pin Lemo port
- To reduce radiation, please keep above 2 meters away from the radio station
- Take lighting protection measures when installing antennas

- Change the cable if damaged

Related Regulations

The receiver contains integral Bluetooth® wireless technology. In some countries, the unit can be used without obtaining an end-user license. But in some countries the administrative permissions are required. For license information, please consult your local dealer.

Use and Care

The receiver can withstand the rough treatment that typically occurs in the field. However, the receiver is high-precision electronic equipment and should be treated with reasonable care.

Warning and Caution

An absence of specific alerts does not mean that there are no safety risks involved. A Warning or Caution information is intended to minimize the risk of personal injury and/or damage to the equipment.

WARNING- A Warning alerts you to a potential risk of serious injury to your person and/or damage to the equipment, because of improper operations or wrong settings of the equipment.

CAUTION- A Caution alerts you to a possible risk of damage to the equipment and/or data loss.

Warranty Notice

ComNav Technology does not warranty devices damage because of force majeure (lighting, high voltage or collision).

ComNav Technology does not warranty the disassembled devices.

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1. Product Description

The M100X GNSS receiver is a new generation of high-precision GNSS receiver independently developed by ComNav Technology Ltd. for high-precision vehicle-mounted positioning and orientation applications. It can realize positioning and orientation functions on a single machine, and can provide high-precision and high-dynamic position, speed, attitude and other parameters in real time. Built-in MEMS inertial navigation sensor, GNSS+IMU fusion computing technology, can easily cope with complex obstruction environments such as urban canyons. All-metal design, IP67 waterproof and dust proof level, can be used in various harsh environments.

1.1. Appearance

The appearance of the M100X integrated navigation receiver host is shown in the figure below.



Figure 1.M100X GNSS Receiver

1.2. Interface and Indicator light

Here introduce the interface and indicator of the M100X.



① **GNSS1:** Main station GNSS antenna (positioning) cable interface.

② **GNSS2:** Slave station GNSS antenna (heading) cable interface.

③ **26PIN cable**: 26PIN data interface, including power supply, GNSS COM, system COM, CAN, network port ,USB and PPS.

④ **4G** Card slot.

⑤ **Power indicator light:** If the red light is on for a long time, indicating that the power is on, and the light is off for a long time, indicating that the power is not on.

⑥ **IUM indicator light**: If the indicator light is off, it means the inertial navigation is not turned on. If the red light is always on, it means the inertial navigation is turned on. If the blue light flashes, it means the initialization is successful but not calibrated. If the blue light is always on, it means the initialization is successful and the calibration is successful.

⑦ **The satellite indicator lights of the master and slave stations**: Flash several times in red every 5 seconds, indicating satellites number of the master antenna Tracks; the blue light flashes several times in blue every 5 seconds, indicating satellites number of the slave antenna Tracks.

⑧ **4G data link indicator light**: The orange light flashes quickly to indicate 4G communication is in progress.

1.3. Wiring Instructions

This section introduces the connection methods of the product. The connection between each accessory and the host is shown in the following figure:

The master/slave antennas are connected to the two TNC RF terminals of the receiver through cables. The receiver is powered and communicates with other devices through the M100X data cable and network cable.



Figure 3. M100X wiring diagram

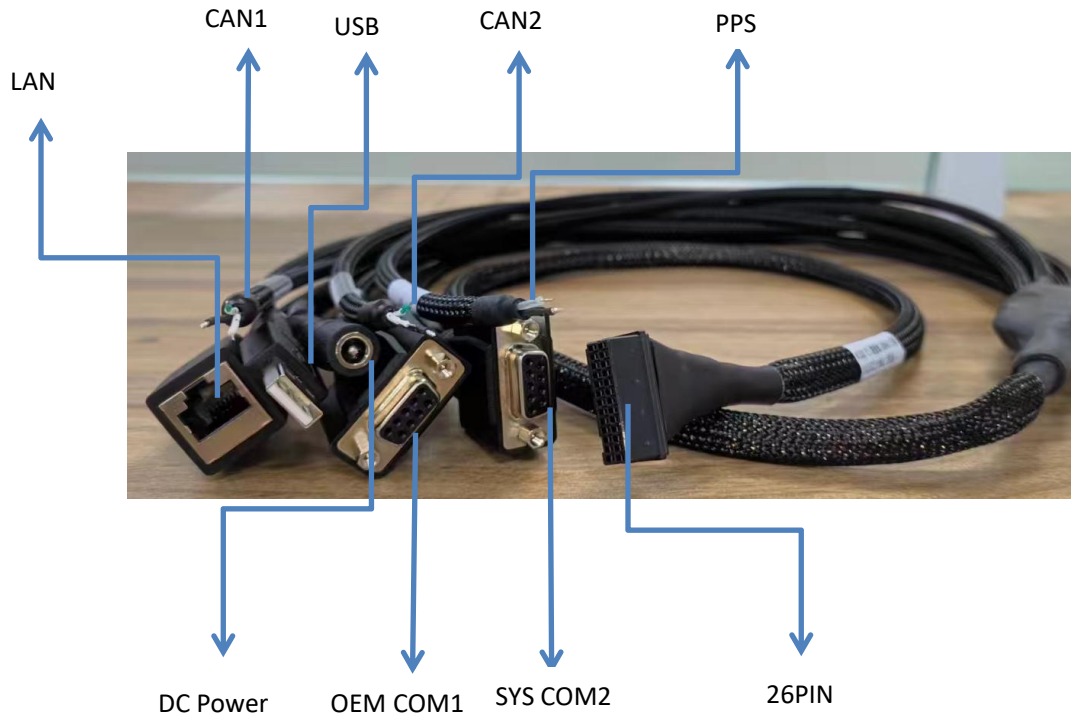


Figure 4. 26Pin cable

1. Install the satellite antenna: After fixing the main satellite antenna (positioning), insert the male end of the cable into the female socket of the receiver satellite antenna interface GNSS1. It is firmly plugged in when you hear a click. After fixing slave antenna (heading), insert the male end of the cable into the female socket of the receiver satellite antenna interface GNSS1. It is firmly plugged in when you hear a click.

2. Install the 4G card: Insert the 4G card into the 4G card slot.

3. Connect the 26-pin data cable: Match the protrusion on the 26-pin data cable male interface with the 26-pin female interface of the receiver and insert it.

4. Connect the power supply: After the 26-pin data cable is connected to the receiver, connect the DC power on the data cable to the external power supply or battery, and the receiver will start automatically after powering on. (Note that the external power supply voltage must be DC 9V~36V)

5. Connect to a computer or integrator:

1) Network port connection: Connect both ends of the network cable to the network port of the receiver and the network port of the computer or integrator respectively, and set the computer IP and the receiver IP to the same network segment;

2) Serial port connection: After the 7-pin data cable is connected to the receiver, connect the COM serial port DB9 female connector on the data cable to the serial port DB9 male connector of the computer or integrator;

3) USB connection: Connect the USB port of the data cable to the computer to access the storage of the receiver;

4) CAN port connection: Connect the DB9 female connector of the CAN port of the data cable to the CAN port of the integrator.

2. Installation

Please read the installation instructions in this section carefully before installing the M100X integrated navigation receiver to avoid affecting the positioning results.

2.1. Receiver Installation

You can use the four mounting angles on the M100 X receiver to mount it on the carrier. The size of the mounting holes is Φ 6.5mm.

Note: When you need to use the IMU function, please pay attention to the installation of the receiver. The device needs to be installed horizontally with the carrier. The IMU settings are different for different device installation directions. For the corresponding setting methods, please refer to Section 4.1.8.

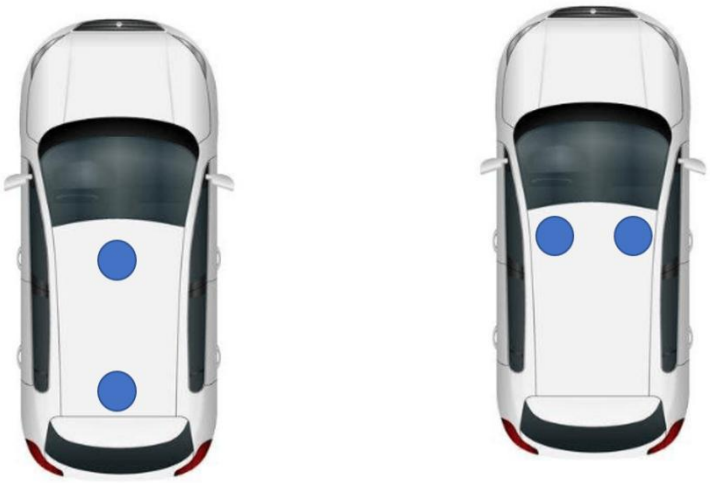
2.2. Antenna Installation

When you need to use the heading function, you need to install two antennas (master and slave antennas) on the carrier and connect them to the receiver. The satellite antennas should face the sky and the carrier should not block the antennas.

You can choose from two installation methods:

Method 1: Please try to keep the connection lines of the front and rear antennas parallel to the direction of the vehicle's travel, with the master antenna at the back and the slave antenna at the front.

Method 2: Please try to keep the connection lines of the left and right antennas perpendicular to the direction of the carrier's travel, with the master antenna at the right and the slave antenna at the left. The master antenna is connected to the M100X GNSS1 port for positioning; the slave antenna is connected to the M100 X GNSS2 port for orientation.



Method 1 Method 2

Figure 5. Antenna installation diagram

After install the receiver and antennas, you please have a check of the indicator light on M100X.

1. Check if the PWR LED is solid red.

2. Check if the SAT LED is flashing red/blue.

Then you can configure the receiver parameters through the Android APP Navigate Master and check the receiver status.

3. Receiver Settings

M100X GNSS receiver can be configured in two ways:

1. Navigate Master software
2. Serial port settings

Please choose the configuration method that suits you according to the usage environment and method.

3.1. App setup

Navigation Master APP is a mobile phone software based on high-precision GNSS to realize navigation applications. It supports the status viewing and parameter configuration of the M100X receiver. Users can download the latest installation package and related technical information from our official website: *https://www.comnavtech.com/*

3.1.1. Bluetooth connection

1. Turn on Bluetooth.
2. Search device: Display available devices normally as shown in the figure.
3. Connect device: Click on the device in the list to initiate a connection.

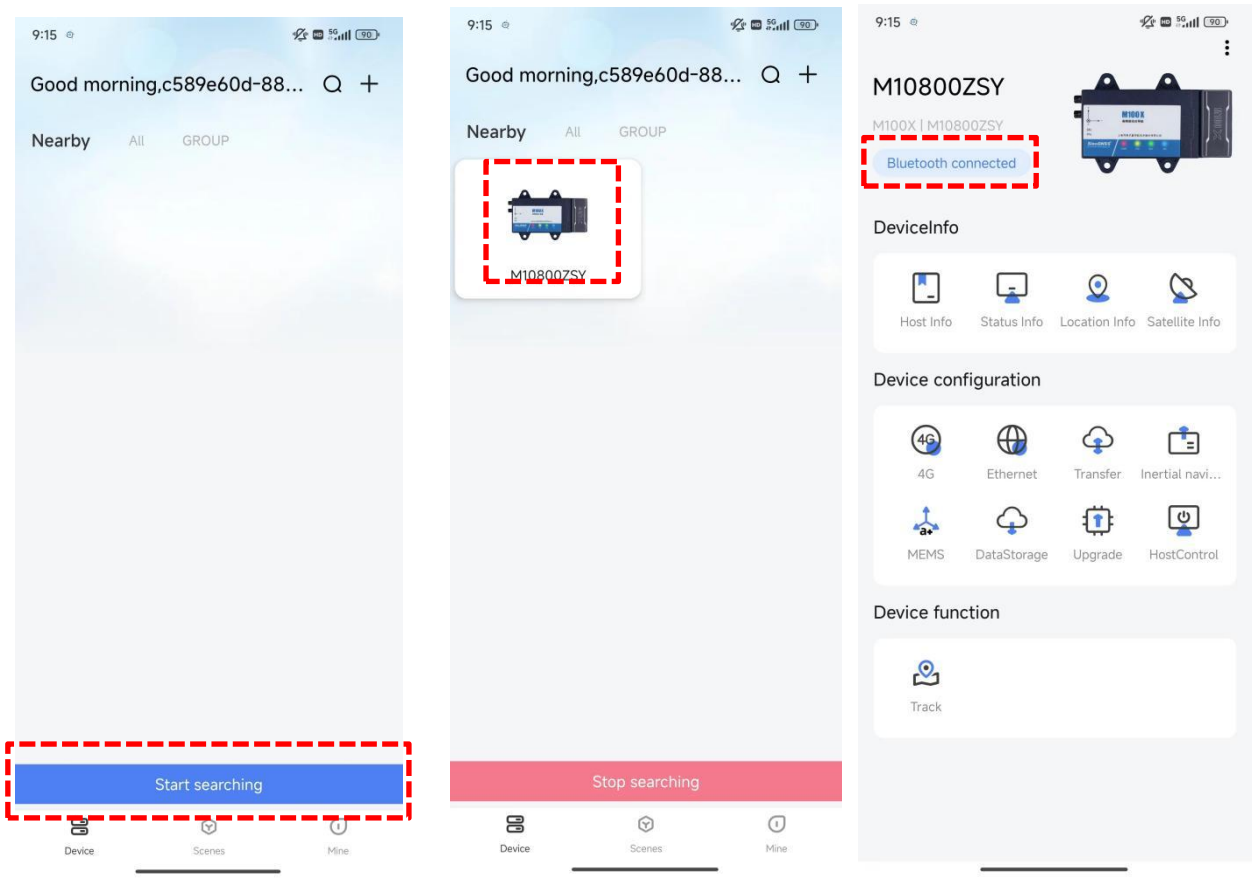


Figure 6. Bluetooth connection

3.1.2. Receiver Information

1. Click [Host Information].

2. Query the device model, device SN, system firmware, board firmware, inertial navigation drive and other information.

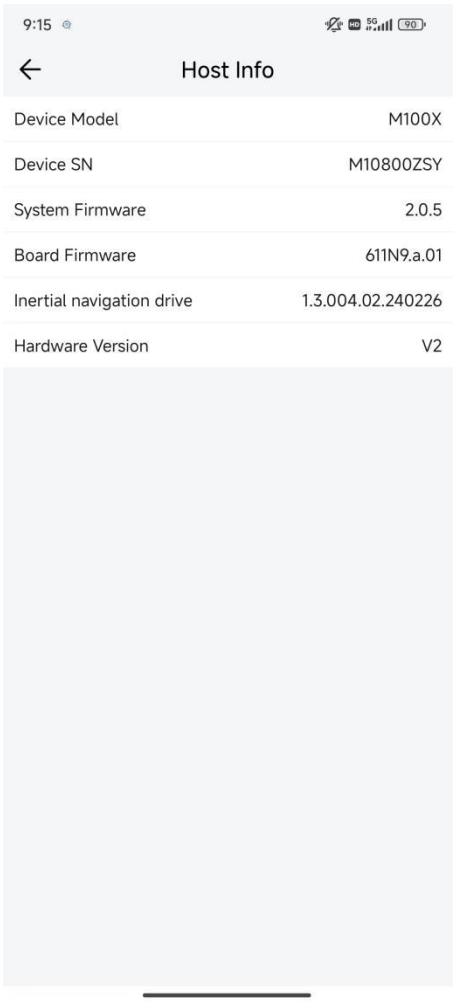


Figure 7. Host information

3.1.3. Location information

Click [Location Information]: Display the receiver clock, location information, antenna signal quality and DOP. The normal display is as shown in the figure.

Note: If the antenna signal quality is less than 80%, the signal quality is poor; 80%-84% is fair; 85%-89% is medium; 90%-94% is good; 95%-100% is excellent. If the signal quality is less than 55%, the antenna is not plugged in.

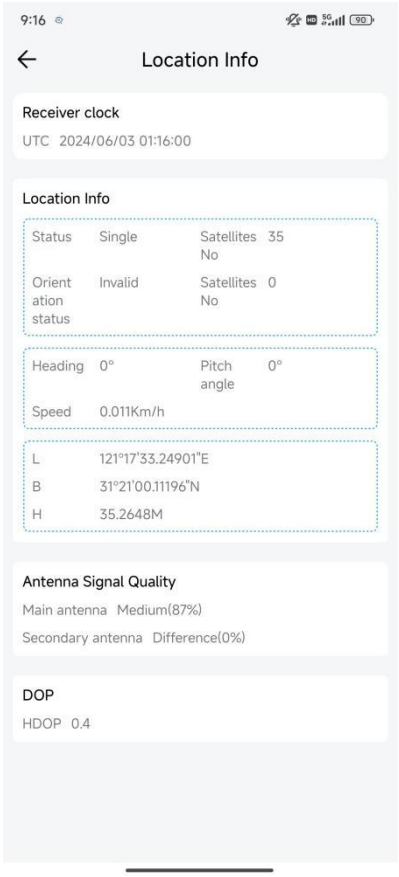


Figure 8. Location information

3.1.4. Satellite information

Click [Satellite Information] to view the satellite status:

Sat is the number of the searched satellite (G for GPS, C for BDS, R for GLONASS, E for GALILEO, and S for QZSS). Azi is the azimuth Ele is the altitude, L1/L2/L5 is the signal-to-noise ratio of this frequency, and B1C/B2a is the BD3 proprietary frequency.

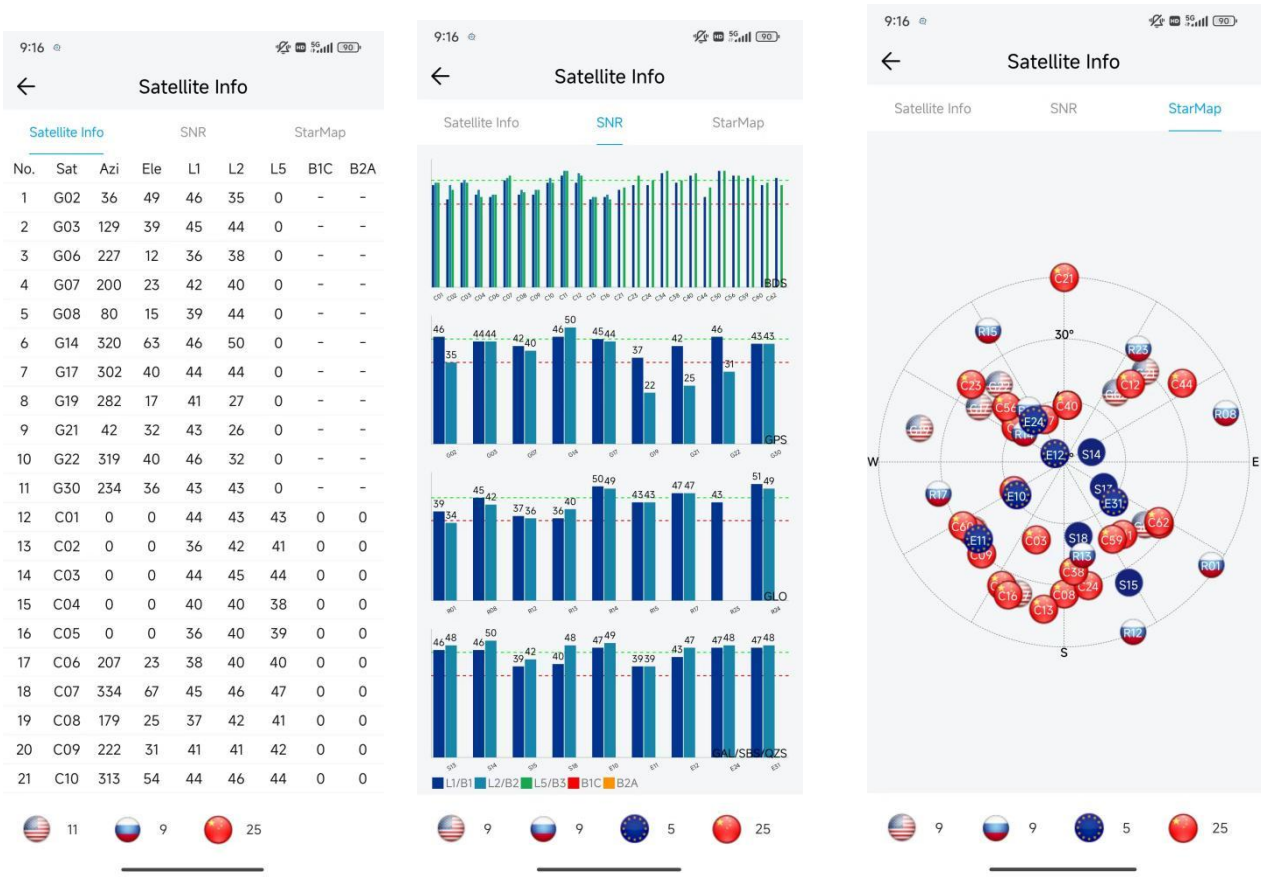


Figure 9. Satellite information

3.1.5. 4G Configuration

1. Click[4G].

2. Turn on the mobile network [switch] and tap Confirm.

3. If the 4G connection is normal, the signal strength, SIM card status, and network status will be displayed correctly.

4. Signal strength: When the signal is displayed as 20 to 30, the strength is normal.

Normally when the card is inserted, and SIM card not detected when the card is not inserted. There are two dialing states: 4G dialing success displays success, 4G dialing failure displays failure. If the dialing state is failure, please contact your 4G card operator.

Note: The receiver does not support hot swapping of 4G cards. Please make sure that the 4G card has been inserted before powering on.

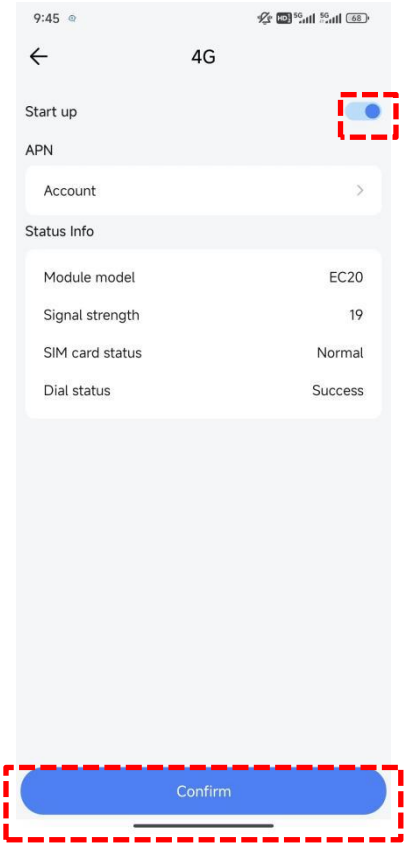


Figure 10. 4G configuration

3.1.6. Ethernet Configuration

1. Click [Ethernet]

2. Configure the IP address, sub net mask and gateway and click OK.

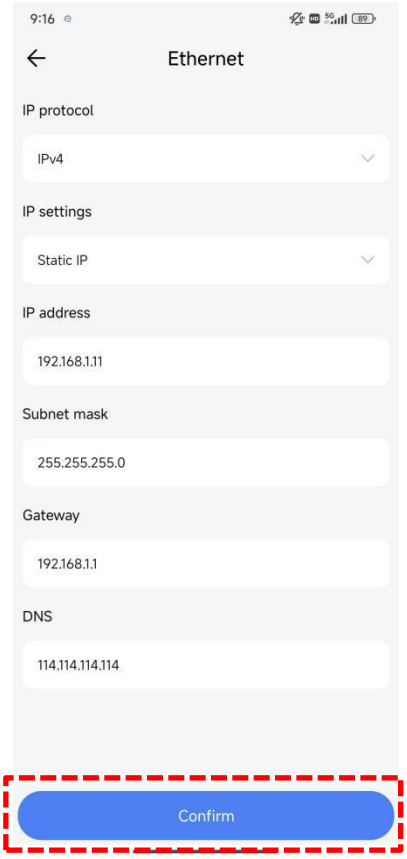


Figure 11. Ethernet configuration

3.1.7. Data transmission

Click [Data Transfer], click [TCP/IP], select Client as the working mode, configure the IP address and IP port of the target server, select the required message, and click [Confirm]. The receiver will automatically upload the message data you selected to the target server.

Note: If you configure the output of IMU raw data, no data is output,Please check whether you have connected the positioning antenna. If the device is powered on but not positioned or the satellite is lost, the IMU raw data will not be output by default.

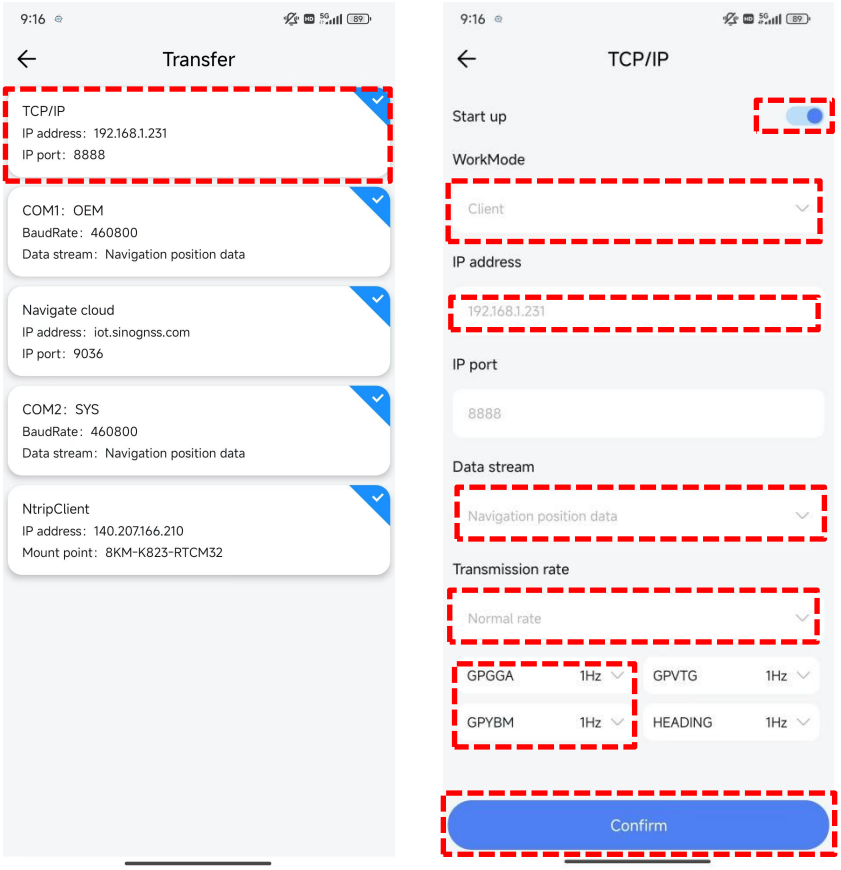


Figure 12. TCP transmission

Click [COM1:OEM], the default baud rate is 115200, configure the baud rate and data stream, and click [Confirm]. Then you can receive the message you configured through the board's 232 port.

Notice:

1. If there is no data output after you complete the configuration, please check whether the baud rate configured on the device is the same as the baud rate of your serial port software.

2. If you send some commands to the board through the serial port tool, these commands will be cleared after the restart. Please reconfigure.

3. If you configure the output of IMU raw data, no data is output:

Please check whether you have connected antennas. If the device is powered on and not positioned or locked, the IMU raw data will not be output by default.

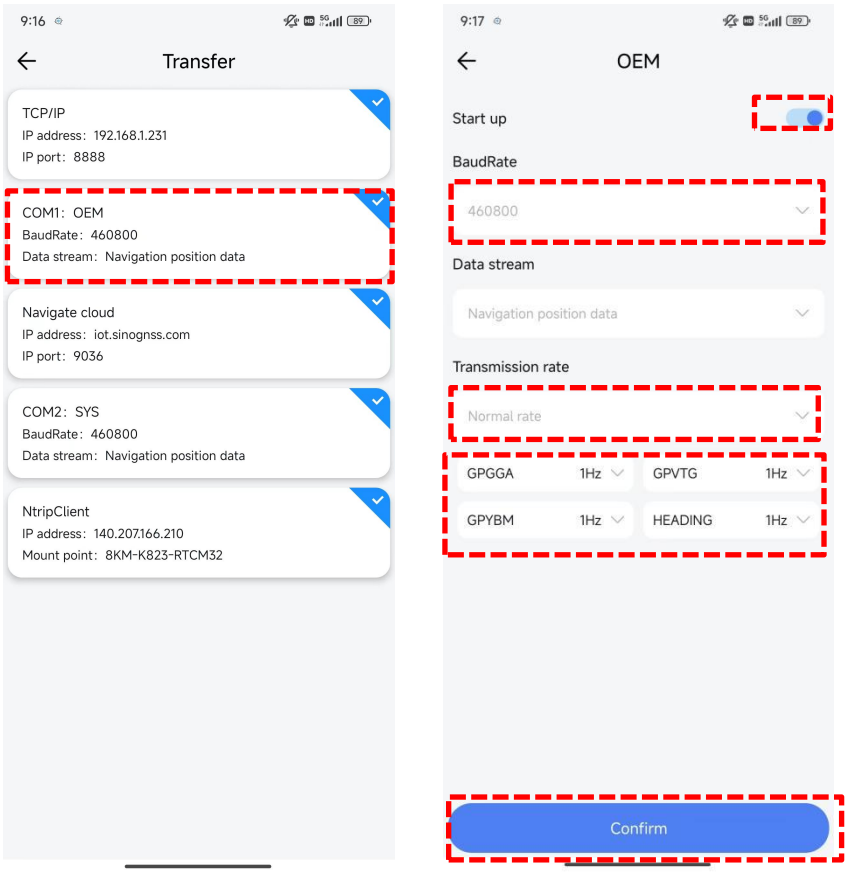


Figure 13. COM1: OEM transmission

Click [COM2:SYS], configure the baud rate, parity check and data flow, and click [Confirm]. Then you can receive the message you configured through the system port 232.

Notice：

1. If there is no data output or no response after you complete the configuration, please check whether the baud rate configured on the device is the same as the baud rate of your serial port software or whether the parity check mode set on your device and serial port software is the same.

2. If you configure the output of IMU raw data, no data is output.

Please check whether you have connected the IMU. If the device is powered on and not positioned or locked, the IMU raw data will not be output by default.

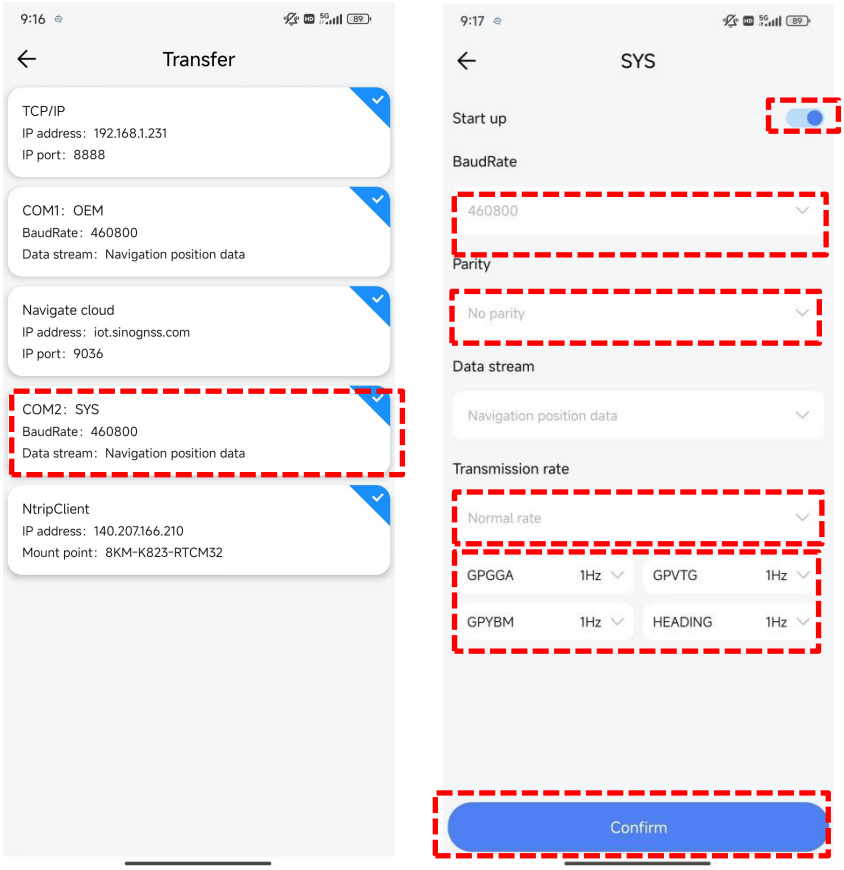


Figure 14. COM2:SYS transmission

Click [NtripClient], configure the differential source IP address, port, user name, password, mount point and other information, and click [Confirm]. Then your device can receive differential data via 4G and output centimeter-level precision positioning data.

Note: If the prompt shows that the setup failed, please check whether the 4G connection is normal.

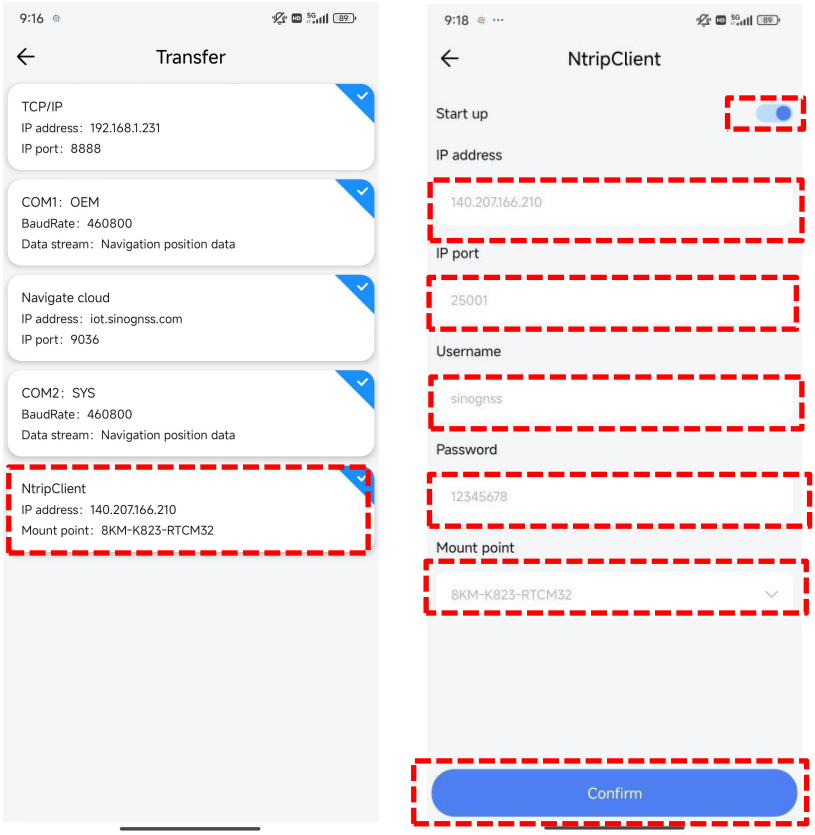


Figure 15. NtripClient configuration

3.1.8. IMU settings

1. Click [INS] to enter the INS configuration page.

2. Set [Initialization Speed]: The initialization speed can be configured according to the speed of your vehicle, but the value must be greater than 1, that is, your vehicle speed must reach 1m/s to complete the inertial navigation initialization. If your application scenario is below 1m/s, you can contact support@comnavtech.com to customize the firmware.

Note: In order not to affect your use, please set the initialization speed reasonably. If the initialization speed you set is too high, your device may not be able to complete the initialization, and may not be able to enter the inertial navigation solution when entering an area with poor satellite signals.

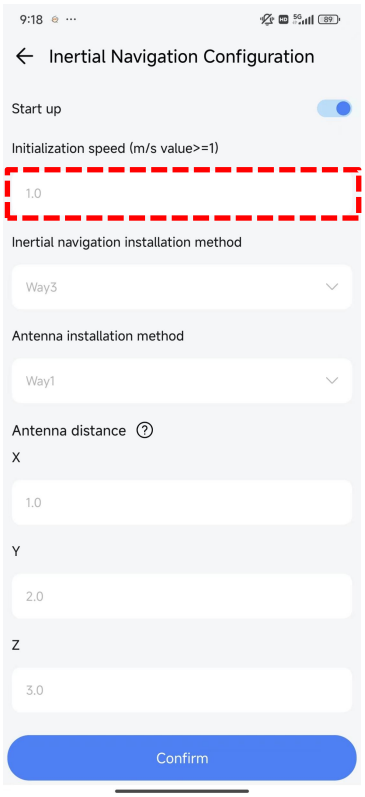


Figure 16. Initialization speed setting

3. Set [INSTALLATION MODE]: You can select according to the actual installation mode of the device.

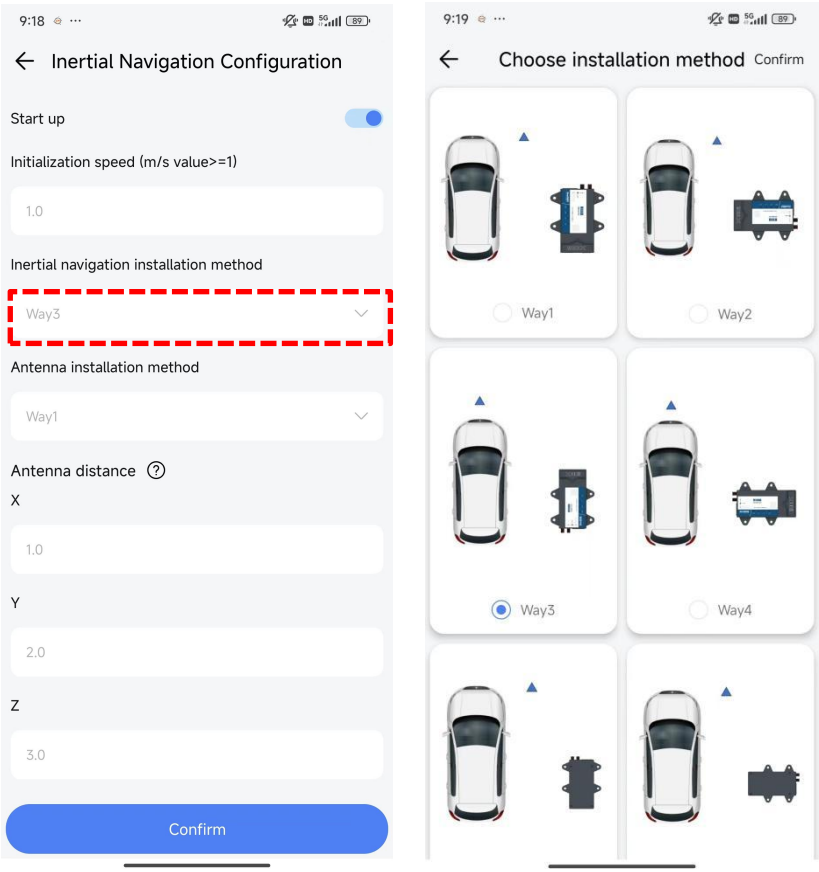


Figure 17. IMU installation configuration

4. In [Antenna Installation Mode], you can choose from two installation modes:

Mode 1: Please try to keep the connection line between the front and rear antennas parallel to the direction of the vehicle, with the main antenna at the back and the slave antenna at the front.

Mode 2: Please try to keep the connection line between the left and right antennas perpendicular to the direction of the vehicle, with the main antenna at the right and the slave antenna at the left. The main antenna is connected to the M100X GNSS1 port for positioning; the slave antenna is connected to the M100X GNSS2 port for orientation.

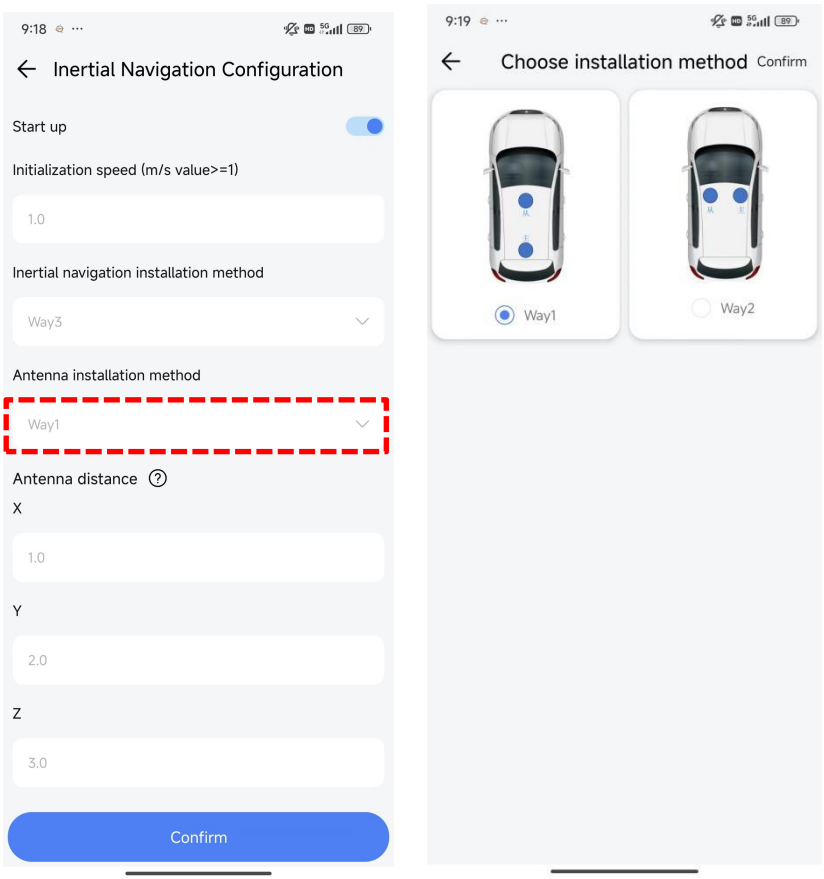


Figure 18. Antenna mounting configuration

5. Set [Antenna Distance], take the inertial navigation device as the coordinate origin, and measure the distance from the positioning antenna (the antenna connected to the device GNSS1) to the device. X is positive when it points to the front of the vehicle, Y is positive when it points to the right of the vehicle, and Z is positive when it points downward. The range of the arm is -20 meters to 20 meters, that is, X, Y, and Z can only be entered between -20 and 20.

Notice：

1. In order not to affect the accuracy of the combined navigation, it is recommended that you control the error within 0.05 meters when measuring the distance.

2. After configuring the inertial navigation parameters, the board module will restart and reset, which is normal.

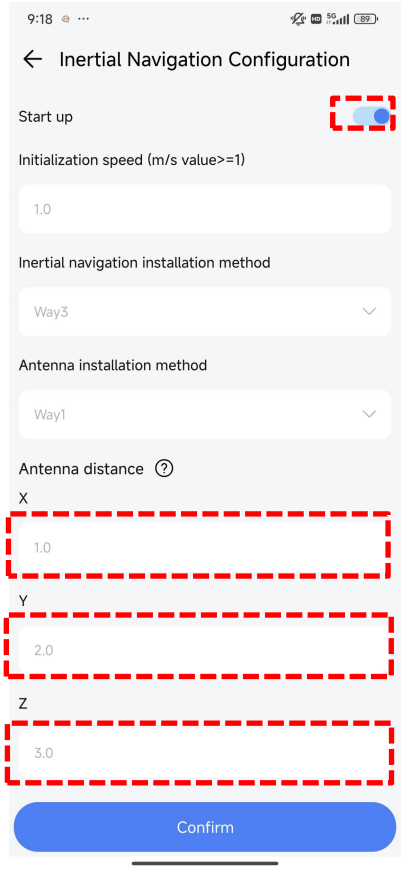


Figure 19. Antenna distance setting



Figure 20. Schematic diagram of the antenna distance measurement origin

After you complete the inertial navigation settings, you need to initialize the device. The initialization method is to connect the differential data to make the RTK enter a fixed solution, drive to an open environment, and drive at a speed greater than the initialization speed you set for more than 10 meters to complete the initialization.

In order to ensure the performance of the inertial navigation system, the device needs to be calibrated after initialization to eliminate the installation error of the device. The calibration method is to maintain a speed of more than 5m/s in an open environment for at least 200 meters, turn left and right twice at 90 degrees (you can make an 8-shaped turn if there is a good road environment), and after the calibration is completed, the inertial navigation light of the device will be blue and steady.

Note:

1. During the initialization and calibration process, please ensure that the environment is open and the RTK is a fixed solution.

2. If your device loses power during driving, you need to complete the initialization and calibration process again.

3.1.9. MEMS

1. Click[MEMS]。

2. In this interface, you can view the three-axis acceleration and three-axis angular velocity of the inertial navigation device in real time.

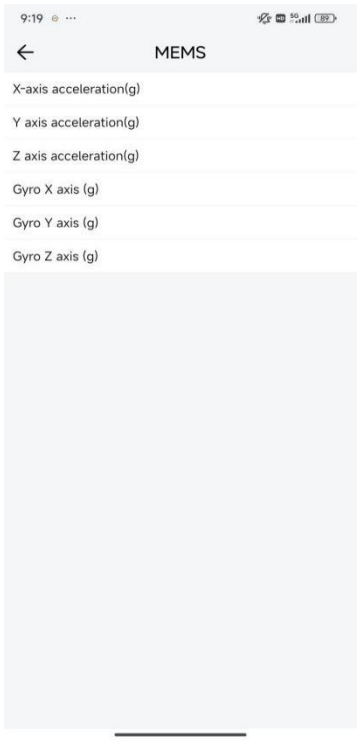


Figure 21.MEMS

3.1.10. Data storage

1. Click [Data Storage] and turn on the [Start up].

2. Set [Storage Space]: You can define it yourself, but the storage space for several records cannot exceed the device storage space. When the storage space you set is exceeded, the device will delete the records from the previous date to the next date until the remaining storage space meets the data storage needs.

3. Set [File Interval]: the time interval between each sampling file. There are several file interval time options: 15min, 1h, 2h, 4h, and 24h.

4. Set [Data Format]: IMU Raw and GGA are available.

5. then click [Confirm], and the device will automatically store the data you selected.

Note: To protect storage, the device does not store data when the USB is connected to the computer by default. When you are storing data, please do not plug the USB into the computer.

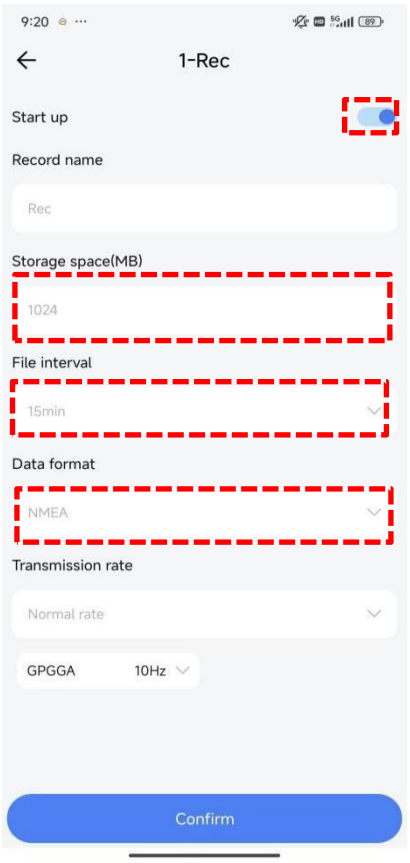


Figure 22. Data storage

3.1.11. Firmware Upgrade

1. [Upgrade mode]:You can choose to upgrade way

2. [Upgrade module]： You can choose to upgrade the board or system according to your needs.

3. [File source]:You can choose File storage location

4. [Select a document]： You can select the upgrade file as needed.

5. [upgrade]：After clicking on the upgrade, you only need to wait for the progress bar to complete the transmission, and then wait for 2 minutes before reconnecting the device via Bluetooth.

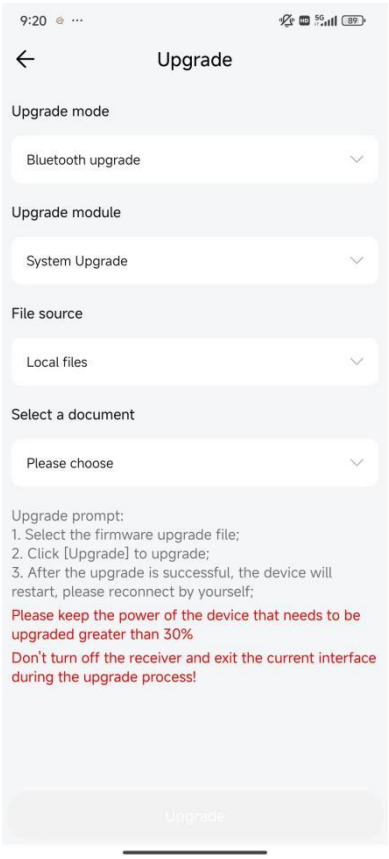


Figure 23. Firmware upgrade

Notice：

1. It takes at least 15 minutes to upgrade the board and at least 5 minutes to upgrade the system. Please do not leave the vicinity of the device during the upgrade process to avoid Bluetooth disconnection and upgrade failure due to incomplete transfer of the upgrade file.

2. Please contact support@comnavtech.com before upgrading to obtain the board firmware or system firmware you need.

3.1.12. Host Control

Operating procedures：

1. [Restart]： After clicking, the receiver will automatically restart.

2. [reset]： After clicking, the receiver configuration will be restored to the factory settings.

Note: After restoring the factory settings, you need to reconfigure the device, so please operate with caution.

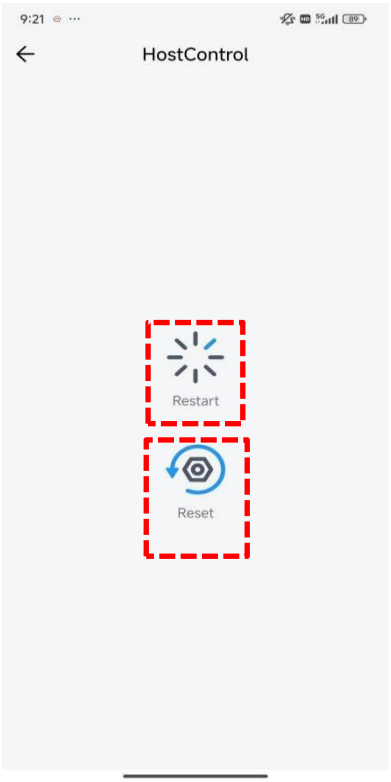


Figure 24. Host control

3.1.13. Trajectory

1. Click [Track] to enter the track work interface.

2. The status bar at the top of the interface can be used to view the current device's satellite search and solution status. When the status displays: Fixed, you can start collecting trajectory operations.

3. Before collecting the trajectory, you need to create a new task and create a task name. After the creation is completed, the APP will automatically record the current high-precision location information.

4. When you have finished collecting, click Stop to complete the trajectory collection.

5. You can click the track button on the right and select the track you just collected in the task list to view the track you collected.

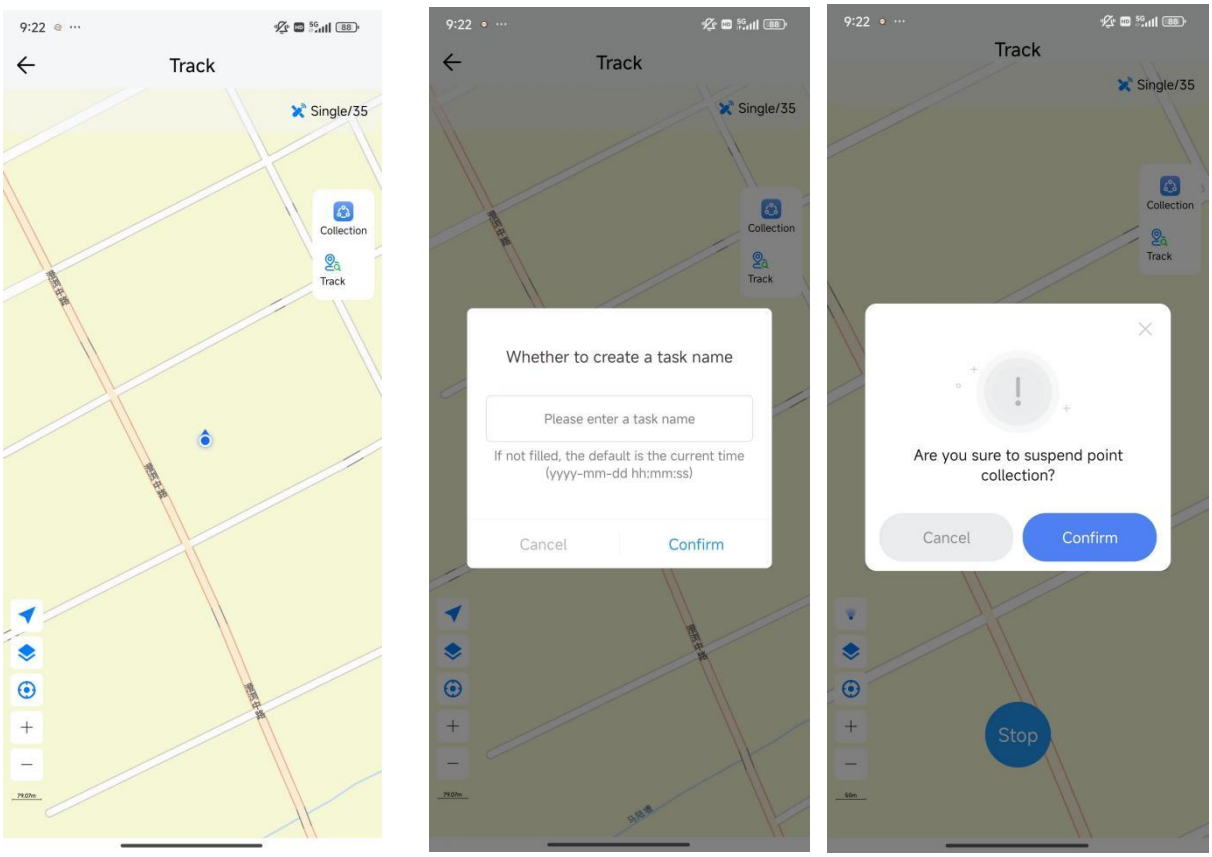


Figure. 25. Trajectory acquisition

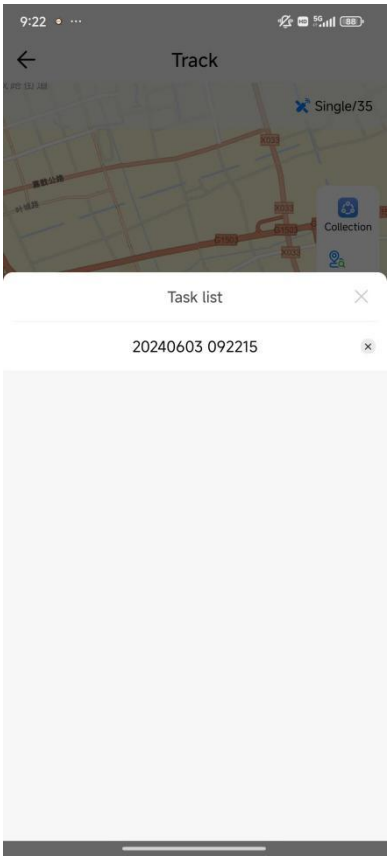


Figure 26. Track View

3.2. Serial port settings

You can also install CRU or serial debugging assistant on a computer and connect M100X com1/com2 to configure it to output NMEA-0183 data. The default baud rate is 115200, and the common commands are:

Board version information：

log version	//check board firmware information

Save Settings：

saveconfig	//save the configurations

Turn on/off inertial navigation：

Inscontrol enable/disable	//enable/disable INS

Set the antenna arm:

Set drleverarm x y z	//setup Lever arm value

Location information：

log gpgga ontime <n>	//n=1,2,5...,The output frequency is 1/n Hz

Direction finding information：

log headinga ontime <n>	//n=1,2,5...,The output frequency is 1/n Hz

Speed information：

log gpvtg ontime <n>	//n=1,2,5...,The output frequency is 1/n Hz

Modify the serial port baud rate：

com com1 9600/19200/38400/115200/460800	// Change the baud rate

Set the IMU post-processing data output port:

Inscontrol enable 1	// 1 Corresponding serial port 1， Set to 0 to not output

Set the inertial navigation initialization speed:

Set drconfig 1.0	//Set the inertial navigation initialization speed threshold to 1m/s

Set the GNSS signal loss inertial navigation recursion threshold:

Set drtimeout 0-3600	//Set the GNSS signal loss inertial navigation recursion threshold

Note: For other setting instructions and message descriptions, please refer to the K8 reference board manual.

4. FAQ

If the problems listed in this section occur when using the M100X GNSS receiver, you can follow the following methods to troubleshoot. If the problem is not solved, please contact our support@comnavtech.com technical support staff in time.

1. The device cannot start

   1) Check whether the power supply voltage is normal. If the power supply is faulty, please turn off the power and replace it with a normal power supply before using the product.

   2) Check whether the power cable is connected correctly and firmly.

   3) Use a Multimeter to measure the connector voltage and check whether the connecting cable is intact. If the cable is faulty, please replace it.

   4) If the above reasons are excluded, the equipment may be damaged. Please contact support@comnavtech.com

2. Serial port not responding

   1) Check whether the serial port number, serial port baud rate, and parity check are correct. Please read the communication protocol carefully to ensure that the serial port can be configured correctly.

   2) Check if the data cable is damaged.

   3) Check whether the serial port driver is installed.

   4) Check whether the data cable is connected reliably.

3. Abnormal positioning

   1) Check if the antenna is blocked.

   2) Check the antenna connection to see if it is secure and correct.

   3) Is the connectivity of the communication system equipment reliable?

   4) Is there any radio frequency interference from the navigation antenna frequency?

   5) Check whether the base station coordinates deviate too much from the actual coordinates.

   6) If the above reasons are excluded, the equipment may be damaged. Please contact support@comnavtech.com.

4. Poor inertial navigation performance.

1) After entering the combination, the positioning track quickly deviates greatly. Please check whether the device installation method is the same as the installation method you set through the APP.

2) After entering the combination, the shorter route has obvious deviation. Please confirm whether you have entered an obstructed environment shortly after initialization and the device has not completed calibration. You can check whether the inertial navigation has completed calibration by checking the inertial navigation light or using GPING messages.